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Research report

Brain imaging of chill reactions to pleasant and unpleasant sounds

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ABSTRACT

The term 'chill' refers to a short-term bodily event of high arousal, which marks an emotional peak experience when occurring in response to music. Chill responses arise in a clearly circumscribed time frame and can also be reliably elicited by unpleasant sounds. Previous research, however, mostly focused on individually selected stimuli and positive contexts, thus, limiting the scope of interpretation. Hence, we developed a standardized chill paradigm and used fMRI to test neural responses of 16 healthy volunteers to pleasant and unpleasant emotional sound material while collecting subjective reports of chill intensity and skin conductance response data. As predicted, we found chill-associated increases in autonomic arousal regardless of the valence of the sound material. Apart from activity in primary and higher auditory cortices, both pleasant and unpleasant chills were associated with anterior insula, thalamus and basal ganglia activity. In contrast, amygdala responses were observed only in association with chills elicited by unpleasant sounds. Thus, chills elicited by pleasant and unpleasant sounds share activity in a neural network that may be specifically involved in the arousal component of an emotional experience.

1. Introduction

A chill is a discrete bodily event of high arousal that is often accompanied by psychophysiological changes such as skin conductance increase or goose bumps and accompanying sensations like shivers or tingles down the spine [1,2]. Furthermore, chills can be felt in the upper spine, neck, shoulders, and scalp and are occasionally accompanied by a lump in the throat, weeping, sighing, or a palpitation [2,3]. Chills in response to music are seen as markers of emotional peak experiences [4]; Panksepp, 1995; [2] and, thus, entail a subjective feeling component and physiological changes as well as cognitive appraisal [5]. Chill passages and no-chill-passages of musical pieces differ regarding the subjective intensity of a feeling as well as the physiological arousal as measured with skin conductance response [6]. The corresponding emotion percept can further be indicated and expressed by the subject and these reports have a high association to psychophysiological parameters [5]. Therefore, studying the chill response provides insight in the processing of emotional states with high intensity of arousal and the additional benefit that the timeframe during which the feeling state rises and subsides is exceptionally clear compared to other emotional states (3–4 seconds [7]).

However, the underlying brain network of chill responses has not been thoroughly investigated. Previous studies used individually selected highly pleasurable music while measuring changes in blood level dependent response with positron emission tomography [8]. The same group [9] later applied dopamine-PET combined with psychophysiological measures of autonomic nervous system activity using again individually selected highly chilling music. To examine the time course of dopamine release, they used functional magnetic resonance imaging (fMRI) with the same stimuli and listeners. They found endogenous dopamine release in the striatum at peak emotional arousal during music listening. In a later partial least square analysis of fMRI-data these authors detected that nucleus accumbens activity and its connectivity strength measured during listening to unfamiliar music pieces predicted the amount of money participants were willing to spend on the corresponding piece afterwards using an auction paradigm [10].

In an ALE-analysis Koelsch [11] identified a number of brain areas found in studies addressing emotional response to music. These comprised the bilateral hippocampus and amygdala, right accumbens, bilateral auditory cortex, left caudate nucleus and the dorsomedial prefrontal cortex (PFC). Other areas showed subthreshold activation: cingulate cortex, orbitofrontal cortex (OFC), left anterior insula, mediodorsal thalamus, superior parietal lobe (SPL; BA7). Most of these brain areas, for example hippocampus, amygdala, ventral striatum,

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caudate nucleus, cingulate cortex, OFC, anterior insula, and thalamus, are also implicated in the chill experience using participant-selected music as shown by Blood and Zatorre [8] and Salimpoor et al. [9].

So far, however, an objective paradigm for evaluating the neural representation of chills elicited by music is lacking. The previously preferred method to induce chill responses is comparing participant-selected music to other music stimuli inducing no chills [8,9,12]. Although this method offers the possibility to evoke emotional episodes with music and gain a high chance of chill events, this paradigm has some flaws. Firstly, this procedure depends on the participant's awareness about individual chill-inducing music. Another problem is that the stimuli are highly familiar to the participants and the effects of familiarity and chill might interact. For instance, it has been shown that emotion-related limbic and paralimbic regions as well as the reward circuitry are more active for familiar than for unfamiliar music [13]. Additionally, memories associated with the familiar music might also have an effect on the emotion experience [14]. The recall of memories can for example be associated with enhanced activity in the striatum [15]. It is impossible to distinguish these influences post-hoc when trying to examine the associated brain regions during a chill passage. The aforementioned studies limited these effects via rigid preselection of participants and stimuli but this hinders the expansion of the chill paradigm to a broader subject group. For instance, a standardized paradigm to examine chills and the underlying brain network is a prerequisite when examining patients who have impairments in chill response due to lesions of the corresponding network [16] or a loss of dopaminergic transmitter [17]. Therefore, when it comes to investigating impaired emotional processing, e.g. in patients suffering from mental or neurological disorders, the high agreement between subjective experience and objectively measurable physiological response as well as the precise definition of the chill concept may help to overcome methodological issues that previously hampered advances in this field of research.

To apply the chill paradigm more generally in the current study, we therefore used experimenter-selected music pieces to induce chill events. These standardized music stimuli have already been found to reliably induce chills and associated psychophysiological responses [14]. As chill responses are markers of intense arousal in general [6,12,18] they are not necessarily pleasant [19]. Just remember the scream going through the classes when the teacher scratches chalk with high frequencies over the blackboard. Investigations at physiological and subjective levels show that apart from tactile stimulation the most potent chill inducers are sounds with negative valence, being even stronger than music with positive valence [5]. Thus, we additional included harsh sounds in our paradigm to ensure a reliable elicitation of chill responses.

This study aimed at (1) developing an objective paradigm for investigations of pleasant emotional chills using standardized music stimuli and evaluating underlying brain networks for regions of interest (ROIs) described to be involved in emotional processing of music and chill response and (2) investigating unpleasant chills with respect to underlying neural representation for the same ROIs. Based on findings by Grewe et al. [5] we expected that chills towards harsh sounds could be evoked more reliably than chills in response to music. In addition, we hypothesized that the former would be experienced more intensively compared to the latter. Since chills reflect highly arousing states, passages associated with chills should be accompanied by larger skin conductance responses compared to passages that comprise no chill reports regardless of chill valence. During chills towards music we expected to observe activity in brain regions associated with reward, emotion processing and arousal (e.g. ventral striatum, thalamus, cingulate, insula, orbitofrontal cortex, dorsomedial prefrontal cortex and cerebellum) as similarly found by Blood and Zatorre [8]. For brain activation during unpleasant chills we expected a similar pattern since there is certain evidence of an U-shaped relationship between valence of acoustic stimuli and changes in brain activity in various regions [20]. We therefore investigated a group of healthy students with fMRI in a study design applying six classical music excerpts and six harsh sounds especially apt for inducing pleasant and unpleasant chills while recording felt chill intensity and skin conductance response continuously.

2. Material and methods

2.1. Participants

We investigated 16 students aged between 18 and 32 years (12 women) recruited from the University of Greifswald. Mean age was 22.3 years (*SD* 3.4). All participants were right handed. Exclusion criteria were self reported mental and neurological disorders, regular intake of medication and impaired hearing. The majority of students reported to have enjoyed further musical education outside school (n = 13) and to play an instrument (n = 11). Whereas rock music was reported as preferred style of music by half of the participants, classical music was named by one participant only.

The study conforms to recognized standards as defined in the Declaration of Helsinki. The study had been approved by the Ethics Committee of the German Society of Psychology. All participants provided written informed consent and received course credit.

2.2. Auditory stimuli

Six classical music excerpts and six harsh sounds which have been successfully applied in past work [5,14,16] were played to induce pleasant and unpleasant chills (see Tables 1 and 2 in the Supplementary Information). Passages in the music excerpts which are especially apt for evoking pleasant chills [14,21] are also listed in Supplementary Table 1. These passages had a length of 7.2–11.5 seconds and contained features such as sudden entrances of solo instruments as well as unexpected tonal and dynamic shifts which are known to be responsible for pleasant chills [22]. The harsh sounds had a length of 7.5–9.9 seconds and were digitally embedded in further excerpts of classical music in that way that they coincided with dynamic and tonal changes representing musical climaxes (see Supplementary Table 2). In previous yet unpublished work we found that using music as an acoustic context enhances the susceptibility for unpleasant chills towards harsh sounds. All acoustic stimuli had a total duration of 90 s. Furthermore, for all stimuli amplitude normalization and fade-in/out effects were applied. Hence, harsh sounds and chill related passages in the music stimuli did not differ regarding volume. Editing of the stimuli was realized with Adobe Audition CS3.0 (Adobe Systems Inc., San Jose, USA). Sound files were stored and played using WAV format. Stimulus presentation was realized with Presentation software and delivered via MRI-capable headphones (MR-confon, Magdeburg, Germany). In a short debriefing after the experiment all participants reported that the applied music excerpts were unknown to them.

2.3. Paradigm and experimental procedure

Before fMRI-measurement, participants received detailed information about the experimental procedure and signed the informed consent form. After attachment of skin conductance electrodes participants were instructed about the upcoming MRI session in which they should listen carefully to the auditory stimuli and indicate each chill experience and its intensity by pressing a foam-covered handle device located in the dominant hand. The degree of the applied force should correspond to the felt intensity of the chill. A chill experience was described as having shivers down the spine, goose bumps or a lump in the throat. By doing this, we covered both approach- (pleasant) and avoidance-related (unpleasant) chill sensations [19]. Afterwards the participants were placed in the MRI scanner and equipped with headphones. At first, anatomical scans were realized lasting about 4 min. Then the twelve stimuli were presented with the constraint that no more than two stimuli of the same category (music, harsh sound embedded in music) appeared subsequently. The volume of the played stimuli was individually adjusted to assure comfortable listening [4]. After each stimulus the sound of white noise was played with a length of 30 s as baseline condition. Single piano tones were embedded in this baseline phase and subjects were instructed to press the device when hearing these tones to control for device pressing in later analysis steps. Throughout the entire experimental procedure, a fixation cross was presented via a projector onto a mirror attached to the head coil. Participants were told to keep their eyes open and to focus the fixation cross in order not to fall asleep and to prevent head movement. After the experiment, participants were repeatedly asked to press the handle device with varying force (25, 50, 75 and 100 % of maximum force) to obtain reference values. Prior to the experimental session in the MRI participants completed a questionnaire regarding musical education, affective reactivity towards music and preferred style of music. The whole experiment took about 45 min.

2.4. Questionnaire and recording of chill experiences during fMRI-measurements

The first part of the questionnaire asked for musical training and whether participants played an instrument. Furthermore, participants were asked to name their preferred styles of music to control for familiarity with the applied stimuli. They were also asked to rate how often they experience certain bodily responses (tears, laughter, lump in the throat, goose bumps, urge to move, shivers, palpitation, trembling, sweating, flushing) towards music on a five-level scale (1 = never, 2 = rarely, 3 = occasionally, 4 = often, 5 = always) as similarly done by Sloboda [2].

The foam-covered handle device used to indicate the intensity of each chill experience was connected via a Respiration Belt MR transducer with a BrainAmpExG MR device. Data recording was then realized with BrainVision Recorder software (all Brain Products, Gilching, Germany) in arbitrary units at a sampling rate of 5000 Hz. Skin conductance was measured using two Ag/AgCl electrodes with a diameter of 4 mm and filled with a 0.05 sodium chloride electrolyte medium [23]. They were placed adjacently on the hypothenar eminence of the palmar surface of the non-dominant hand at a distance of 15 mm. Connected to a GSR-MR module and BrainAmpExG MR device the signal was amplified and recorded with BrainVision Recorder software (all Brain Products, Gilching, Germany). Sampling rate was set at 5000 Hz.

2.5. MRI measurement

Images were collected with a 3 T Magnetom Verio (Siemens, Erlangen, Germany) using a 32-channel head coil. During the run, 774 volumes with 33 slices (1 mm gap) were acquired using echo-planar imaging (EPI; TR 2000 ms; TE 23 ms; flip angle 70°; FOV 192 × 192 mm^2; matrix size 104 × 104, voxel size 2 × 2 × 4 mm^3). The first three volumes in each run were discarded to allow for equilibration effect. Thirty-four phase and magnitude images were acquired in the same FOV by a gradient echo sequence (flip angle 60∘; FOV 192 × 192; slice thickness 3 mm; TR 488 ms; TE1 4.92 ms; TE2 7.38 ms) to calculate a fieldmap aimed at correcting geometric distortions in the EPI images. T1-weighted structural scans were also acquired (MP-RAGE, TR 1690 ms, TE 2.52 ms, flip angle 9°, matrix size 256 × 256, voxel size 1 × 1 × 1 mm^3). The whole experiment took about 45 min.

2.6. Chill experience reports

Chill experience data (recorded via the handle device) were processed with BrainVision Analyzer 2.0 (Brain Products, Gilching, Germany). A built-in peak finder routine was applied to look for signal peaks. The peaks were then treated as chill experiences and their values as reported intensity. To cope with slight pressure changes over time the intensity values were corrected for the baseline pressure of the corresponding section. Finally, the obtained intensity values were divided by the participant's largest reference grip strength recorded in the separate session after the experiment. In doing so, individual differences regarding grip strength were compensated. To examine the chill evoking capability of the applied music and sound stimuli, analyses of chill frequencies were focused on chills during harsh sounds (unpleasant chills) and during preselected passages in the music stimuli especially apt for evoking chills (pleasant chills; [14]). Absolute numbers of unpleasant chills were divided by the length of the harsh sound stimuli resulting in chills per minute values [18]. Analogously, absolute numbers of pleasant chills were divided by the length of the preselected music passages. A paired *t*-test was computed to compare the means of chills per minute values for pleasant chills during the preselected chill passages and unpleasant chills during harsh sounds. Spearman's rank correlations were calculated between the absolute numbers of pleasant chills and the frequency ratings of the bodily responses associated with music. For the analyses of chill intensity only those participants were considered who reported both unpleasant and pleasant chills. A paired *t*-test was computed to compare their mean intensities. For the correlations and analysis of intensities all pleasant chills towards music were considered (hence, also chills before and after the harsh sounds). Chill intensity data of two participants were not considered for analysis since reference grip strength values were not available for them.

2.7. Processing and analyses of skin conductance data

Skin conductance data were preprocessed using BrainVision Analyzer 2.0 (Brain Products, Gilching, Germany), down-sampled to 10 Hz and exported to a txt-file. Further calculations were then realized with Ledalab software package [24] in Matlab which separates the signal into a tonic and phasic component through a deconvolution approach. For all analyses SCR was scored as the average phasic driver in μS which constitutes the most adequate measure of phasic activity [24]. Also, following Venables and Christie [23] a logarithmic transformation was carried out to normalize the distribution. SCRs associated with chills during music was examined using two strategies.

In an individual approach the average phasic activity was calculated for a time window of 10 s around each chill towards music starting five seconds prior to the chill to capture previously described rise and fall times of the chill response.

When chill reports followed at a distance of less than 10 s only the first chill was used for analyses. For comparison we calculated the average phasic activity during 10 s windows randomly chosen from the same music stimuli, which did not coincide with a chill. These windows did not overlap with the chill-associated 10 s windows or with an occurrence of a harsh sound. Data from 15 participants were considered for psychophysiological analyses since one participant reported no pleasant chill at all. Since we expected a chill-associated increase in SCR, an one-tailed paired *t*-test was applied to compare mean values associated with a chill with mean values not associated with a chill [25].

As individual responses can be biased by individual levels of responsiveness, in a second approach analysis of SCR was focused on the preselected passages in the stimuli irrespective of the participants' chill report (predefined approach). Average phasic activity was calculated in a 9 s window starting one second after the onset of the preselected music passages (see Table 1). Similarly, average phasic activity was also calculated in a 9 s window starting one second after the onset of the harsh sounds.

Following the approach applied by Guhn et al. [14] mean values were computed across all trials in which a chill was reported separately for harsh sounds and the preselected music passages. Also, mean values were computed for all trials in which no chill was reported. A two-way analysis of variance (ANOVA) with Stimulus (preselected chill passage

Table 1
Frequency of chill reports and mean chills per minute (CPM).

Musi			Harsh sounds		
	% of participants with at least one chill	CPM preselected chill passage		% of participants with at least one chill	CPM
Albinoni	62.5	–	Styrofoam1	68.8	6.68
Barber	62.5	–	Styrofoam2	68.8	6.14
Bruch	68.8 (31.3)[1]	1.95	Fingernails	56.3	5.34
Chopin	62.5 (25)	1.49	Fork/Porcelain	62.5	5.91
Mendelssohn	62.5 (37.5)	3.9	Fork/Claypot1	56.3	5.5
Mozart	62.5 (18.8)	2.08	Fork/Claypot2	43.8	3.42

[1]preselected chill passage in brackets.

in music vs. harsh sounds) and Chill (chill report vs. no chill report) as variables was conducted to compare changes in SCR associated with chills with changes in SCR not associated with chills for both chill conditions (music and harsh sounds). Furthermore, an analysis of covariance (ANCOVA) was computed with Stimulus (preselected chill passage in music vs. harsh sounds) as variable and reported chill intensity as covariate to examine differences in SCR associated with chills while controlling for their reported intensity. Statistical analyses of chill reports and skin conductance were carried out using Statistical Package for the Social Sciences (IBM SPSS Statistics 22, 2014). Nominal degrees of freedom and effect sizes are reported. Alpha was set at .05.

2.8. Processing and analyses of fMRI data

FMRI data were analyzed using the SPM12 software (Wellcome Department of Cognitive Neurosciences, London, http://www.fil.ion.ucl.ac.uk./spm) running under Matlab (Math Works; Natick, MA). Spatial pre-processing included realignment to the first scan and unwarping of geometrically distorted EPIs using the Field Map Toolbox. Due to some setting error 10 subjects had aliasing artifacts in the T1 anatomical images. Therefore, functional images were normalized to Montreal Neurological Institute (MNI) space using the SPM EPI template instead of using the anatomical images. Normalized functional images were smoothed using the smoothing function in SPM12 with a Gaussian kernel of $8 \times 8 \times 8$ mm^3 full-width at half-maximum to increase the signal to noise ratio and reduce inter-subject differences. Similar to SCR, BOLD responses were examined using two different strategies (individual and predefined). First, the BOLD responses during the pleasant chills were individually modeled for each subject using the time points of peak device pressure as reference for chill moments (individual approach). In the general linear model, a time window of 10 s around each chill event served as time window of interest. The onset of the window was defined 5 s before the chill. Pleasant chills during music were only modeled individually if a participant had more than one device pressure event during the paradigm (2 subjects did not). If another chill event took place within 5 s after a chill only the first one was modeled. We discarded events with device pressure amplitude of less than 0.5 % of maximum amplitude (2 % of all subjects' events) and events which occurred in the first 6 s of the experiment as this might rather be a startle or orientation response (1.5 % of events). Those parts of music stimuli without chill events were modeled separately for comparison. Unpleasant chills were also modeled individually depending on device pressure and covered the whole stimulus duration of up to 10 s. Again sound stimuli without chill device pressure were modeled separately. The white noise was included as implicit baseline to control for holding the device and unspecific listening. For all first level analyses, realignment parameters were inserted as additional regressors and a high pass filter of 128 s was applied as suggested in the default settings.

Group level analyses comprised one-sample-t-tests for each of these conditions (music with individual chill reports, music without chill reports, unpleasant sound with chill reports, unpleasant sound without chill reports) compared to the implicit baseline, as well as one-sample-t-tests for the comparison (music with chill reports vs. without chill reports).

In a second approach (predefined approach) BOLD responses during pleasant chills were modeled using the preselected chill passages in the music stimuli known to contain potent chill-inducing features [14]. A window of 10 s starting with the onset of the preselected passages served as the applied time window of interest. Music stimuli around the preselected passages were modeled separately. Unpleasant chill events were modeled in an analogous manner covering the whole stimulus duration of up to 10 s regardless of individual chill reports. The white noise was again included as implicit baseline.

Group level analyses comprised one-sample-t-tests for each of the conditions of interest (predefined chill passage, harsh sound) compared to the implicit baseline. We used a region of interest (ROI) approach correcting for multiple comparisons within the selected ROI using family-wise error (FWE) with $p < .05$.

Based on the ALE analysis by Koelsch [11], and previous chill experiments by Blood and Zatorre [8] and Salimpoor et al. [9], we expected the following ROIs to be involved in the neural response to chill-inducing stimuli: auditory cortex, anterior insula, hippocampus, amygdala, nucleus accumbens, caudate nucleus, dorsomedial PFC, cingulate cortex, OFC, and thalamus. Some of these regions have been only correlated negatively (hippocampus, amygdala) with chill intensity when using pleasant stimuli [8] but might be strongly implicated in the processing of unpleasant stimuli [20,26,27]. In addition, the cerebellar hemispheres have been described to be important for autonomic response and intensity of experienced emotional response to different emotional stimuli [28] and have also been positively correlated with chill intensity [8]. As there are several areas, especially motor areas, that might show up during the chill experiences but are not of interest here, we decided to only focus on specific regions that have been shown previously to be associated with emotional processing of music and chill experiences in particular. Therefore, we used the following ROIs ($n = 22$; 4 medial areas were not differentiated between hemispheres) from the Anatomy atlas: hippocampus, amygdala, primary auditory cortex (A1) and higher auditory cortex (A2-A3), and thalamus. From the AAL we applied the following masks: anterior cingulate, caudate and putamen, and cerebellar hemispheres. From the Harvard-Oxford-atlas we used masks for the OFC and the nucleus accumbens. The anterior insula mask was taken from Neuromorphometrics and those of the dorsomedial PFC was differentiated by the Sallet atlas ([29]; cluster 3; BA 9). To illustrate the activation differences related to pleasant chills further we also extracted the percent signal change of three regions of interest (primary auditory cortex, thalamus, anterior insula) in each hemisphere using the RFX toolbox [30]. Focusing on coordinates given by the combined contrast of music with and without chills we extracted the activation in an 8 mm sphere for each participant in both conditions and compared these activation levels via paired Wilcoxon tests in R.

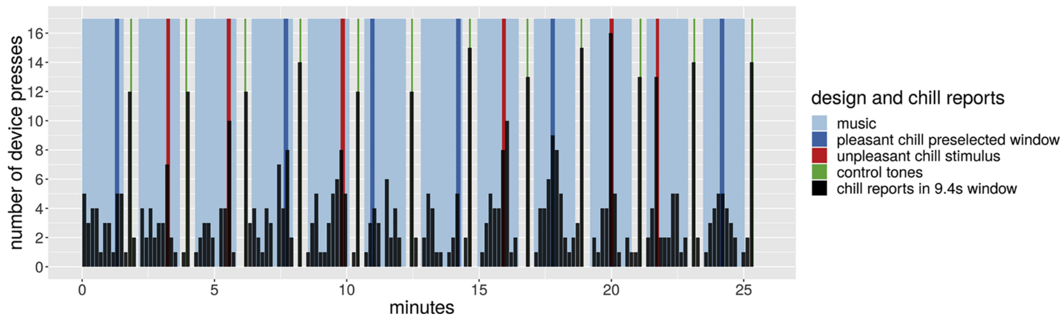

Fig. 1. Course of the experiment in minutes with music (light blue) and harsh sounds embedded in music (red). Music passages especially apt for evoking chills are shown in dark blue. Absolute numbers of chills are illustrated as black bars. Between the stimuli white noise featuring single piano tones was played (green).

3. Results

3.1. Chill reports

While at least one unpleasant chill was reported by 13 participants (81.3 %), one pleasant chill or more was reported by 15 participants (93.8 %) during the music excerpts. Although the preselected chill passages in the music stimuli constituted only a small proportion of the applied music material half of the participants reported at least one chill during these passages. Playing harsh sounds resulted in a higher number of chills (chills per minute) compared to the preselected music chill passages ($t(15) = 3.308$, $p = .005$, $d = .827$; see Table 1 for number of chills for each participant and left panel of Fig. 2). The participants' rating about the frequency of music shivers was positively correlated with the reported number of pleasant chills ($r = .643$, $p = .010$). Unpleasant chills were reported slightly more intense compared to pleasant chills reaching marginal significance ($t(10) = 1.833$, $p = .097$, $d = .553$). Fig. 1 provides an overview on chill reports and stimulus design of the whole experiment.

3.2. Skin conductance data

As expected, results of the ANOVA showed that experiencing a chill was associated with a higher increase in SCR compared to the absence of a chill during music and harsh sounds resulting in a significant Chill factor ($F(1, 156) = 14.860$, $p < .001$ and $\eta^2_p = .087$; see right panel of Fig. 2). Also harsh sounds evoked a more pronounced increase in SCR compared to the music passages (Stimulus factor: $F(1, 156) = 8.872$, $p = .003$ and $\eta^2_p = .054$). No significant interaction between the two factors was found (Chill x Stimulus: $F(1, 156) = 0.976$, $p = .325$ and $\eta^2_p = .006$). Single comparisons supported the finding of chill-associated increases in SCR (preselected chill passage in music: $t(62) = 1.836$, $p = .041$, $g = .667$; harsh sounds: $t(94) = 3.891$, $p < .001$, $g = .738$). Also, the individual approach showed that SCR associated with chills during music was significantly larger compared to SCR when no chills were reported during music ($t(14) = 2.89$, $p = .006$, $d = .747$). While controlling for reported chill intensity SCR was larger during harsh sounds with chills than during preselected music passages with chills indicated by a significant Stimulus factor ($F(1, 59) = 8.234$, $p = .006$ and $\eta^2_p = .122$).

3.3. fMRI data

We uploaded t-maps of the most important comparisons to the neurovault repository (https://identifiers.org/neurovault.collection:6084).

3.3.1. Individual approach

Both pleasant and unpleasant chills were accompanied by activation in several of the hypothesized auditory and emotion processing areas. In comparison to the implicit noise baseline especially primary and higher auditory cortices were activated (Tables 2A and 3A). Bilateral anterior insula, thalamus, putamen, caudate nucleus, and cerebellar hemispheres were activated in each condition. Only in the unpleasant chill condition right OFC and bilateral amygdala showed activation.

Comparing the individual pleasant chill windows to music without chill events we found only minor activation differences in auditory areas but profound differences in bilateral anterior insula, thalamus, putamen and left caudate nucleus (Table 2B). Further differences could be found in right cerebellar hemisphere. Fig. 3 shows the activation differences in these areas and illustrates in particular the implication of the anterior insula in the pleasant chill experience along with averaged percent signal changes in some of the reported areas. When comparing music with chill events (left) and music without chill events (right) activation in primary auditory cortices showed no difference (left: V = 60, p = .6698; right: V = 49, p = .8552). In contrast, activation within the thalamus and anterior insula showed substantial differences

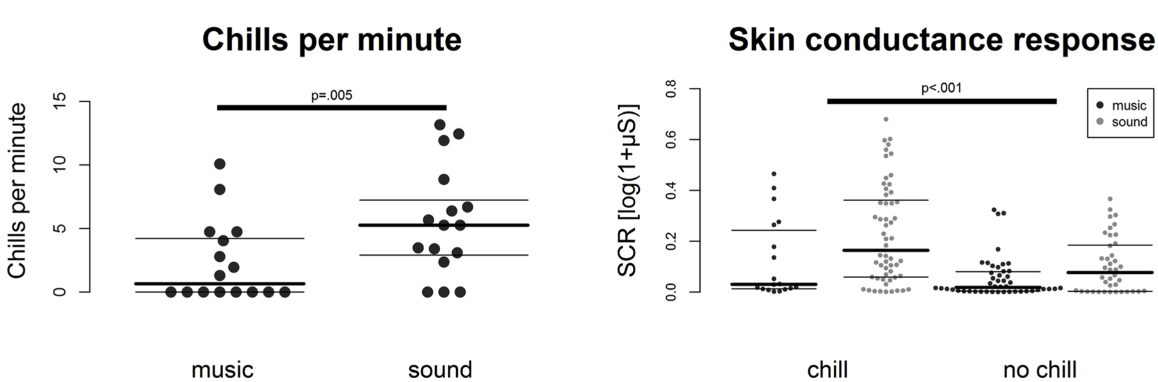

Fig. 2. Swarm plots showing individual mean chills per minute values for preselected music passages and harsh sounds (left panel) and each skin conductance response for preselected music passages and harsh sounds with and without chills (right panel). Thick horizontal lines represent median values, thin horizontal lines represent quartiles.

Table 2A
MNI coordinates for the contrast music with individual chill > white noise baseline.

Pleasant chill						
Anatomical area	x	y	z	t-value	p(FWE[3])	Cluster
Primary auditory cortex L[1]	−50	−14	0	11.53	< 0.001*	276
Primary auditory cortex R[2]	58	−4	−3	12.55	< 0.001*	282
Higher auditory cortex L	−58	8	0	9.88	< 0.001*	272
Higher auditory cortex R	60	−4	−6	13.62	< 0.001*	487
Anterior insula L	−32	18	9	6.63	0.002*	189
Anterior insula R	42	10	0	6.44	0.003	165
Thalamus L	−18	−14	9	6.43	0.006	93
Thalamus R	22	−10	6	5.83	0.012	25
Putamen L	−26	−2	6	7.32	0.001*	210
Putamen R	22	10	0	6.45	0.003	225
Caudate L	−20	20	6	6.03	0.005	26
Cerebellar hemisphere L	−22	−64	−51	6.60	0.029	286
Cerebellar hemisphere R	24	−66	−51	6.85	0.020	370

[1]left; [2] right; [3] pFWE corrected for ROI volume; the * marks those which survived further Bonferroni correction for the number of ROIs (n = 22; p < 0.0023).

Table 2B
MNI coordinates for the contrast music with individual chill > music without chill.

Pleasant chill > music without chill Anatomical area	x	y	z	t-value	p(FWE) [3]	Cluster
Primary auditory cortex L[1]	−48	0	−3	4.90	0.018	9
Higher auditory cortex L	−58	8	0	6.29	0.005	40
Anterior insula L	−30	16	9	6.71	0.002*	253
Anterior insula R[2]	44	14	−6	6.34	0.004	223
Thalamus L	−20	−12	9	6.14	0.009	137
Thalamus R	18	−8	0	5.83	0.013	37
Putamen L	−26	14	12	7.37	0.001*	46
Putamen R	24	−6	3	5.47	0.012	35
Caudate L	−20	22	6	6.99	0.002*	39
Cerebellar hemisphere R	16	−52	−21	7.76	0.008	650

[1]left; [2] right; [3] pFWE corrected for ROI volume; the * marks those which survived further Bonferroni correction for the number of ROIs (n = 22; p < 0.0023).

Table 3A
MNI coordinates for the contrast harsh sound with individual chill > white noise baseline.

Unpleasant chill Anatomical area	x	y	z	t-value	$p_{(FWE)}$ [3]	Cluster
Primary auditory cortex L[1]	−36	−32	9	11.88	< 0.001*	264
Primary auditory cortex R[2]	44	−24	3	11.48	< 0.001*	270
Higher auditory cortex L	−66	−36	12	9.30	< 0.001*	409
Higher auditory cortex R	58	8	−6	21.94	< 0.001*	576
Anterior insula L	−32	14	3	11.47	< 0.001*	374
Anterior insula R	36	26	0	12.46	< 0.001*	263
Thalamus L	−18	−6	12	5.72	0.018	40
Thalamus R	14	−2	6	4.94	0.043	76
Putamen L	−28	14	−9	10.62	< 0.001*	159
Putamen R	20	4	12	5.51	0.013	106
Caudate L	−18	−2	15	5.93	0.008	19
Caudate R	16	4	15	5.96	0.007	51
Cerebellar hemisphere L	−34	−60	−51	7.65	0.013	800
Cerebellar hemisphere R	32	−60	−54	6.73	0.030	76
OFC R	30	22	−18	7.20	0.008	48
Amygdala L	−24	0	−18	4.45	0.025	13
Amygdala R	28	0	−21	5.66	0.005	55

[1]left; [2] right; [3] pFWE corrected for ROI volume; the * marks those which survived further Bonferroni correction for the number of ROIs (n = 22; p < 0.0023).

demonstrating the specificity of these structures for highly arousing events (thalamus left: V = 101, p = .0009; right: V = 99, p = .0017; anterior insula left: V = 103, p = .0004, right: V = 92, p = .0107).

Comparing the individual unpleasant chill windows to sounds without chill events, we found only small activation differences in left anterior insula and right caudate (Table 3B).

3.3.2. Predefined approach

In the predefined approach analyses most of the hypothesized areas for pleasant chills showed no activation comparing the preselected music passages with baseline (Supplementary Table 3). When comparing these passages with the surrounding music none of the activations that had been identified in the individual analyses could be found. When comparing all unpleasant sounds (regardless of chill occurrence) with baseline the activations in the hypothesized areas were nearly identical to the ones reported for individual unpleasant chill events (Table 3B).

4. Discussion

In the current study, we used emotional sound material of both valences to induce pleasant and unpleasant chills and investigated reported chill intensity, skin conductance response and the neural representation for chill responses in healthy subjects using fMRI. Increased psychophysiological responses (SCR) were observed for those passages specified as chilling by the participants, thus, indicating a useful paradigm for standardized chill response investigations. Especially the harsh sounds reliably evoked chills of high felt intensity, which were also associated with increased autonomic arousal as measured with SCR.

Furthermore, a number of regions showed to be involved in chill response processing. Both pleasant and unpleasant chill stimuli were processed in primary and secondary auditory cortices as well as in anterior insula, thalamus, putamen, caudate and cerebellar hemispheres. Comparing unpleasant chill stimuli to baseline additionally lead to activation in bilateral amygdala and right OFC. It was demonstrated using direct comparison between music stimuli with and without chill that most of these activations were highly specific for the chill response. Especially areas responsible for modulation of arousal and emotional appraisal are heavily involved in the chill experience, as it could be shown by activation in anterior insula, thalamus, putamen, and caudate.

Due to possible differences in responsiveness, we also pursued a second approach using preselected time windows for the investigation of pleasant chills, which have been shown to be effective for chill induction. We observed quite different results for this evaluation approach than with the individual approach leaving only a marginally activated set of ROIs (Suppl. Table 3) and no activation specific for the difference between these preselected passages and the rest of the music. Therefore, the individual approach might be especially suited for an individual phenomenon like the pleasant chill experience.

For unpleasant sounds, the direct comparison of sounds with chill events to sounds without chill events led to only minor activation differences in left anterior insula and right caudate. We also additionally pursued the predefined approach using all harsh sounds regardless of a chill occurrence and observed nearly identical activations to what we observed when comparing the individual unpleasant chill events with baseline (compare Table 3A and Supplementary Table 4). Due to the strong potential of unpleasant sounds to evoke chills, there is a lack of comparable unpleasant but non-chilling sounds. Although this suggests, that an individual approach might not be necessary for unpleasant sounds, the chill-specificity can be discussed controversially (see limitations).

Music stimuli led to pleasant chill events in many participants underlining the stimuli's potential to induce chills as shown before [14]. Furthermore, since none of the participants reported familiarity with

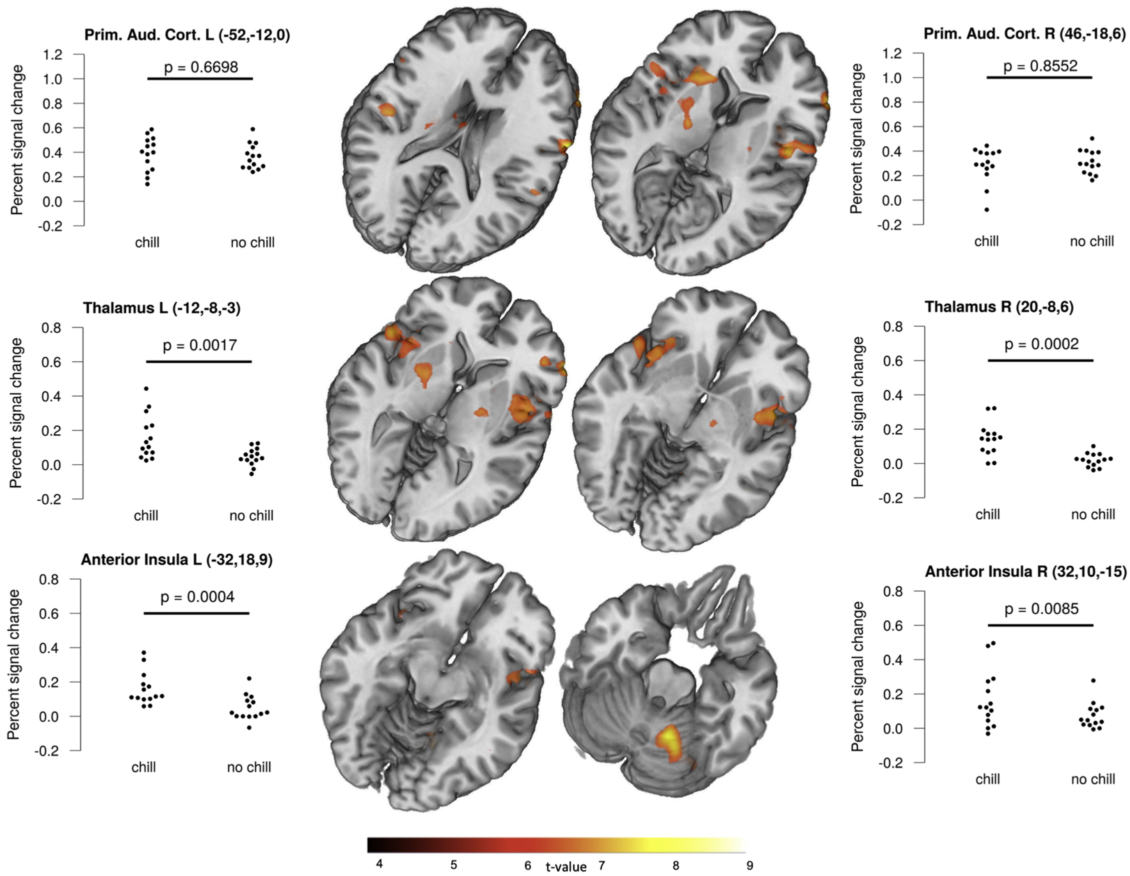

Fig. 3. Comparison between music without and with chill events (individual approach); middle: fMRI activation map contrasting these conditions individually, group activation map projected on Collin's brain with a threshold of $p \leq 0.001$ sliced from top to bottom with $z = 20, 13, 9, 0, -10, -20$; left and right: percent signal changes in swarm plots in specific regions, comparing music with chill events and music without chill events showing no differences in the auditory cortices but profound differences in thalamus and anterior insula demonstrating an activation of a specific network only during chill events.

Table 3B
MNI coordinates for the contrast harsh sound with individual chill > harsh sound without chill.

Unpleasant chill > sound without chill						
Anatomical area	**x**	**y**	**z**	**t-value**	**p(FWE)** [3]	**Cluster**
Anterior insula L[1]	−42	4	−9	5.77	0.040	6
Caudate R[2]	18	22	9	6.18	0.031	4

[1]left; [2] right; [3] pFWE corrected for ROI volume.

the music stimuli and only one participant listed classical music as preferred style of music, the induced chills can be treated as unaffected by individual associations and memories [14]. Interestingly, the number of participants reporting a pleasant chill was similar compared to the study by Guhn et al. [14]. The number of participants reporting unpleasant chills was in line with previous work [5].

A moderate positive linear relationship between self-reported susceptibility for shivers and the actual number of experienced chills towards music supports a recent finding by Colver and El-Alayli [31] who showed that participants can reliably estimate their tendency towards chills.

In line with previous work [5] unpleasant chills towards harsh sounds were associated with an increased skin conductance indexing autonomic arousal. Also in accordance with the majority of studies dealing with chill towards music [6] we found an increase in skin conductance associated with pleasant chills.

We found associations of chill response with fMRI-activation unspecific for valence in the auditory cortex, anterior insula, thalamus, putamen, caudate, and the cerebellar hemispheres. This is in line with research on emotional responses to music [11] and previous research on chill experience using participant-selected music [8,9].

Although unknown music stimuli evoke less chills than known ones [32], they offer the possibility to implement a broadly applicable paradigm, if potent stimuli are used. However, positive valence stimuli such as music showed a highly individual variability in reported chill responses between participants. Therefore, an fMRI-evaluation using predefined onsets showed only partially comparable results to an analysis using individual onsets indicated by participants' experienced chill. The individual approach seems to be particularly useful when investigating such a highly individual phenomenon.

When comparing individual pleasant chill passages with music without chill, the compared stimuli are of the same quality leading to a comparable activation level in auditory cortices. However, striking differences can be detected in areas like thalamus, anterior insula, putamen, and caudate, which are all strongly modulated by a chill experience (see Fig. 3). It is well known that the thalamus modulates arousal dimensions of emotional stimuli (e.g. [33]). The anterior insula has previously been described to be especially active during the processing of unexpected chords [34]. Overall, the anterior insula is directly associated with multimodal emotional stimuli processing [35]. A loss of feeling for music after left AIC lesion has already been described [36] as well as loss of craving for smoking [37]. The specific role of the insula in chill experience has been demonstrated in a study associating the white matter connectivity between auditory processing areas and insula with individual differences in the sensitivity to chills [38].

In the present study, the bilateral anterior insula and thalamus, as well as putamen and caudate, were consistently activated independent of the valence dimension. This supports the component model of emotions by [39] postulating a contribution of different brain regions to the emergence of a subjective feeling: an affective/arousing component (thalamus), a sensory-interoceptive component (anterior insula), a motor component (basal ganglia), and a cognitive appraisal component (OFC). The basal ganglia activation must not necessarily be associated with motor function. In particular the caudate nucleus is interconnected with prefrontal areas functionally related to various cognitive processes [40,41]. Consequently, the caudate nucleus has been found involved in the acquisition of different skills, such as in working memory training [42], and has been implicated in learning of stimulus- response associations [43]. In addition, the cerebellar hemispheres show important function for feed forward processing not only relevant for motor tasks but also for emotional processing of auditory stimuli [44]. Both the basal ganglia circuit and the cerebellar circuit are parallel processes to cortical circuits associated with learning and automatization of motor, sensory but also cognitive function [41]. The caudate has also been mentioned in previous chill research in terms of its role in prediction and context during music listening and evidence is provided that the caudate is involved in the anticipation phases several seconds before chill experiences [9]. As implicit and explicit knowledge about musical structures play a role in anticipation, this area of research is still of great interest [45]. It has been suggested that violations of expectations about the musical structure can lead to chill experiences [22]. The specificity of the anterior insula and caudate activation for pleasant and unpleasant chills in our data further underscores this idea.

It has been shown, that the nucleus accumbens might be implicated in the peak emotional response phase during the time of a chill when using participant-selected music [8,9]. However, listening to familiar music in comparison to unfamiliar leads to differences regarding the brain activation in the reward system [13]. Other listening paradigms with unfamiliar music might show activation in nucleus accumbens [46] but as Pereira et al. [13] also noted this activation was subthreshold. In our data, the experience of chills was not related to nucleus accumbens activity. Previous research showed, that even when chill epochs were excluded from analyses the nucleus accumbens still showed hemodynamic activity which was related to pleasure level [9]. Predictions about upcoming music phases might be more precise and associated with more pleasure when the music is familiar. First studies regarding the topic of familiarity indeed report that increases in self-reported familiarity significantly enhance experienced pleasure via the arousal component of emotion [47], but the basic mechanism might be the same. A more recent study has shown that, regardless of familiarity, arousal levels during music listening are causally linked to fronto-striatal networks and argue that their role in anticipation produces the emotional reaction [48]. Although spatial incorrectness induced by preprocessing procedures and artifacts might not allow for a definite answer to the question regarding the role of the nucleus accumbens in the chill experience, our imaging data support previous considerations that chills are physiological markers of intense arousal in general [6,9,18]

In contrast, the amygdala was only active in unpleasant chill responses in line with previous research on unpleasant sounds [20,26,27]. We showed that unpleasant sounds are indeed potent inducers of chill experiences. The accompanying increase in arousal as measured with SCR was more pronounced compared to chill reactions towards music stimuli even when controlled for reported chill intensity. An explanation could be that the applied sounds showed a high level of acoustic roughness which is characterized by temporal fluctuations of intensity [49]. It was shown that roughness is an essential feature of human and artificial warning sounds resulting in an increased bilateral amygdala activity and inducing aversive states in the listeners which is interpreted as threat processing [50]. Hence, one could conclude that chills towards harsh sounds also mirror aversive states in the context of warning behavior as it was similarly done in previous work [19,51]. In this respect, a high number of participants reporting chills towards harsh sounds appears plausible since adequate behavior towards threat is relevant for survival [50]. Another explanation could be the higher general intensity of the used unpleasant stimuli, so that a definite answer regarding the specificity of amygdala activation requires further work. Beside this, playing unpleasant harsh sounds can constitute a reliable strategy to induce chill experiences and should be addressed further.

4.1. Limitations

Our findings regarding pleasant chills should be restricted to Western listeners since there is an ongoing debate about cultural aspects of music processing [52]. Given that knowledge about musical structure is associated with the experience of chills [22] one could claim that participants who differ regarding musical enculturation also differ regarding chill reactivity towards a specific type of music. However, in the era of a globalized music industry and unlimited availability of music the cultural background might become less and less relevant when dealing with emotional reactions towards music.

As in previous studies on this topic ([4]: 65.8 %; [6]: 62.1 %) the number of participants in our study having experienced musical education outside school (81.3 %) is certainly not representative for the general population [53]. To increase external validity in future work we recommend to balance the extent of music training of participants and to explicitly recruit those describing themselves as music laypersons.

Negative valence chill stimuli were so effective in eliciting chills that a comparison on the individual basis between brain images during chill-related and chill-unrelated sounds was underpowered. This is also the reason for the consistent results between the predefined and individual approach. Another condition of aversive but non-chilling sounds should be inserted (e.g. highway sound) as an additional control condition for testing chill specificity for the negative valence stimuli. It would then be of interest to investigate whether negative valence stimuli elicit chills in a less individual manner than the music stimuli, which is suggested by the consistency of results in our two approaches. This would lead to an easier prediction of chill occurrences and the possibility to investigate broader subject groups with fixed paradigms.

We did not compare brain activations for pleasant and unpleasant chill responses directly as the applied pleasant and unpleasant chill stimuli differed regarding their acoustic nature (harsh sounds vs music). However, the stimuli were chosen that way that chills of both valences could be reliably evoked. Since we were primarily interested in chill-associated functional activation and sympathetic arousal separately for both valences we did not match our applied stimuli regarding valence and arousal ratings. Therefore, as similarly noted by Grewe et al. [5] comparisons regarding chill frequency, intensity and SCR should be done carefully.

From this study, we cannot decide whether frequency of chills is actually a function of valence or whether a higher number of unpleasant chills is rather due to the greater intensity of the evoking chill stimuli. Grewe et al. [5] found relations between chill frequency and subjective ratings (valence and arousal) to differ between stimulus modalities (sounds, music). An explanation could be that it is challenging to find acoustic stimuli that cover all quadrants of the two-dimensional affective space equally [5,54]. Matching the stimuli with respect to valence, arousal and intensity will help to increase comparability. In a next step, it might be advantageous to choose pleasant and unpleasant stimuli within the same acoustic domain. As a contrast to pleasant classical music unpleasant music stimuli could be recruited from the genres of experimental music or contemporary classical music that features atonal and noisy themes [55]. In the case of sounds, a variety of stimuli that differ regarding valence and arousal could be used [5,54]. Hence, unpleasant but less arousing and chill-inducing sounds like the background noise of a motorway or clock tick might be thinkable allowing the creation of adequate contrasts.

Applying pleasant and unpleasant stimuli of the same acoustic nature might be better suited to conduct direct comparisons of the evoked chill reactions especially with regard to physiological reactions and BOLD responses. These comparisons will shed further light on the anatomic foundations and commonalities of pleasant and unpleasant chills.

To sum up, we found that regardless of valence, chills were associated with an increase in autonomic arousal (SCR) and activity in the anterior insula, thalamus and basal ganglia. Whereas chills elicited by harsh sounds could be evoked more reliably, pleasant chills in response to music had to be investigated on an individual basis.

5. Conclusions

We successfully introduced a strategy to examine pleasant and unpleasant chills with fMRI by playing classical music and harsh sound stimuli. Despite of the highly chill specific music material, an individualized onset fMRI-analysis was necessary to depict activation in a chill specific brain network. For the harsh sound stimuli, a predefined onset for fMRI-analysis can be used but the chill-specificity of the paradigm as it stands can be questioned. The use of unknown music stimuli and even negative valence stimuli to induce chills opens the horizon for more general considerations whether experienced chills represent an intense arousal associated with characteristic psychophysiological reactions. Further work on patients with selective lesions modulating different aspects of emotional processing might help to understand these issues.

Author statement

Klepzig K.: design of the study, data measurement, data evaluation, and writing of the paper

Horn U.: data measurement, data evaluation, and writing of the paper

König J.: design of the study, data measurement

Holtz K.: design of the study, data measurement

Wendt J. design of the study, data evaluation and writing of the paper

Hamm A.O.: design of the study and writing of the paper, oversight and leadership responsibility

Lotze M.: design of the study, data evaluation, and writing of the paper, Oversight and leadership responsibility, acquisition of the financial support for the project leading to this publication together with Hamm A. O.

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